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Towards Underground Mobile Fleet Electrification

Three Essential Steps to Make a Real Change

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Electrifying haul trucks and other vehicles in underground mines will have a significant impact on reducing greenhouse gas emissions and operational costs and improving worker health and safety.

The Australian mining industry is the sector with the second largest contribution to carbon emissions (about 100 Mt CO_2-e pa) in the country and has the largest growth in emissions over the past 30 years. It consumes approximately 14% of the total Australian energy use. Figure 1 shows the Australian mining industry energy usage for the period between 2015 and 2020 and indicates an annual growth of approximately 2%. The sector obtains roughly 50% of its energy from diesel, 22% from natural gas, and 25% from grid electricity, with the rest provided by a combination of renewables, biofuels, and other fossil fuels.

A significant amount of greenhouse gas (GHG) emissions is associated with underground mining operations. These generally use diesel-powered vehicles and thus require high-power ventilation systems to provide acceptable air quality levels for underground workers.

Using a battery electric vehicle (BEV) fleet will eliminate diesel emissions and hence reduce the exposure of workers to fumes and will also substantially reduce the underground ventilation power requirements. To achieve net-zero emission mining operation, electrifying the vehicles and shifting to renewables is essential.

However, there are significant challenges in electrifying mining fleets:

1. Lower energy density of the batteries versus diesel fuel.
2. Slower recharging time versus diesel refuelling.
3. Limited commercial availability of electric mining vehicles and higher upfront cost of mining BEVs.
4. Increased electric energy demand for use in vehicle battery charging.
5. Retraining the workforce in electric vehicle skills.

These could impact electrification costs and productivity. Therefore, it is essential that the mining industry has the right tools and strategies for informed decision-making toward net-zero emissions operation. Drawing from our work with the Mine Operational Vehicle Electrification (MOVE) project, funded by the Future Battery Industries Cooperative Research Centre (FBICRC) in partnership with BHP Nickel West, IGO and various other mining companies and institutions, we outline the three essential steps for successful electrification. In conclusion, we provide recommendations for the future of mining electrification.

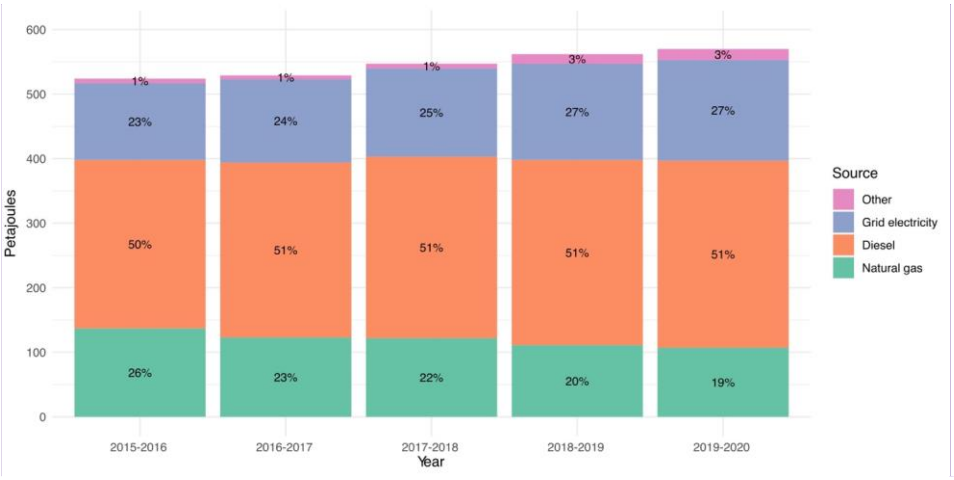

Figure 1: Usage of energy sources by the mining sector in Australia

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Section 1 – Mining Electrification Background

Mining companies have traditionally been risk averse. The uncertainty associated with locating economic ore bodies already poses significant risks, and there is a tendency to avoid additional risks. However, there is increasing pressure from society, governments, investors, and purchasers of mineral products (particularly for the manufacturing of electric vehicles) to reduce emissions. Consequently, most mining companies have committed to substantial emission reduction targets, which can only be achieved through electrification and the increased use of renewable energy. Despite the considerable challenges, there is widespread recognition within the industry that the shift toward mine electrification will occur soon.

Moreover, there is a consensus that broad collaboration within the industry is necessary to facilitate progress towards this goal. According to the 2020 survey of industry executives by State of Play, summarised in Figure 2, mining electrification and renewable energy integration, followed by the

adoption of new technologies, such as automation, will be the main trends for the next 15 years. This is evidenced by recent initiatives and projects launched by the mining industry, as shown in Figure 3.

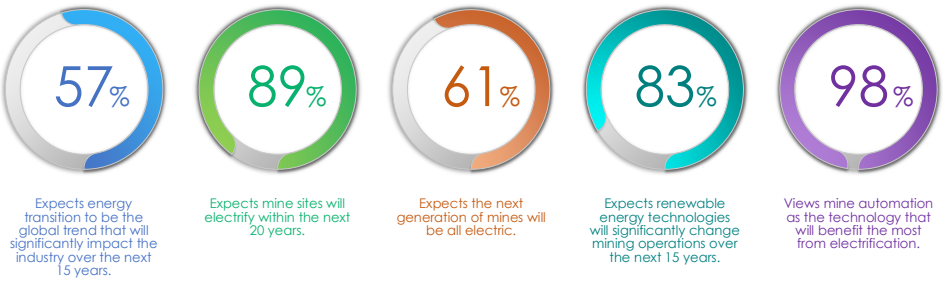

Figure 2: Mining industry executives survey results about the future direction of the industry, conducted by State of Play

Some of the efforts initiated by the sector are briefly mentioned here. The Global Mining Guidelines (GMG) group was established in 2012 to bring the global mining community together to foster collaboration and networking. In 2017, they published a general guideline on mine electrification, which was updated in 2022. This guideline is an excellent source of information on considerations for the electrification of mines. The International Council on Mining and Metals (ICMM) launched the Innovation for Cleaner and Safer Vehicles in 2018, bringing together leading international mining and metal companies and large original equipment manufacturers (OEMs) to accelerate the development of a new generation of mining vehicles and improve existing ones.

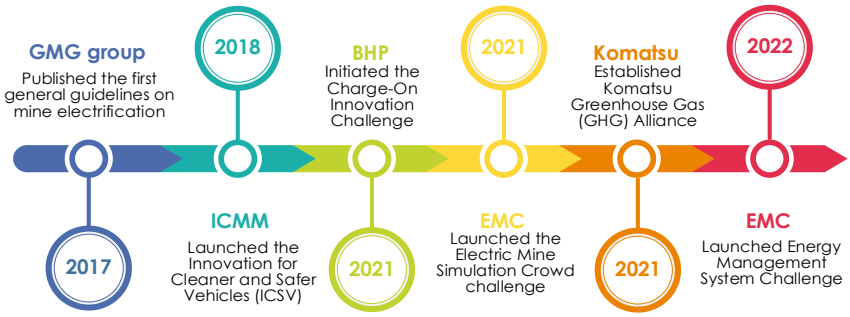

Figure 3: The most recent efforts initiated by the mining industry for mining electrification.

The Charge-On Innovation Challenge, initiated in 2021 and led by BHP, was a global initiative to develop concepts for large-scale haul truck electrification systems to help the mining sector reduce its diesel fuel consumption and significantly reduce emissions from surface mining operations. Several mining companies, service companies, and other industry stakeholders joined this challenge, and the eight winning technology innovators were announced in 2022, including ABB, Ampcontrol and Tritium (Australia), BluVeinXL, DB Engineering & Consulting (DB E&C) and Echion Technologies, Hitachi Energy, Shell Consortium, Siemens and 3ME Technology.

The Electric Mine Consortium (EMC) was formed in 2021 to accelerate technology development and adaptation towards fully-electrified, zero-carbon-emission mining. Several mining companies, service companies, OEMs, battery providers, power system operators, consulting firms, and other industry stakeholders joined this effort. The EMC launched the Electric Mine Simulation Crowd Challenge in 2021, led by OZ Minerals (recently acquired by BHP), to attract companies around the world to propose solutions for developing a scalable electric mine design simulation platform. They also launched the Energy Management System Challenge in 2022, led by South32, whose objective was to bring stakeholders together to create solutions to the energy management systems of mines.

The Komatsu Greenhouse Gas Alliance was established in 2021 to promote collaboration on trialling Komatsu's pre-production vehicles and offering the option to be the first buyers of the solutions once commercially available. The founding members were Rio Tinto, BHP, Codelco and Boliden.

The general consensus among the industry is reflected in several reports published recently, e.g., EY in 2019, VCI and NREL in 2020, which sees mining electrification as a foundational enabler to achieve zero emission mining. This offers significant immediate and long-term benefits across economic, health, and environmental dimensions.

Section 2 – Mining Electrification Main Drivers

The three major drivers for mining electrification are explained in the following subsections.

Subsection 2.1 – Environment, Social, and Governance (ESG) Considerations

There is increasing pressure on nations to agree to global targets for GHG and to be seen as socially responsible citizens. As a critical industry in Australia with a considerable share in total GHG emissions, the mining industry will be closely scrutinised to reduce its emissions.

Social influence is becoming increasingly important for driving global changes. End users of minerals, such as EV manufacturers, are becoming more interested in demonstrating the "greenness" of the manufacturing of their products, given this is an important factor for some of their likely target markets. This mainly affects metals for batteries, such as lithium, nickel, cobalt, and other common metals found in vehicles, such as iron and copper. As a result, Bloomberg News Energy Finance has introduced the

Bloomberg Green's Electric Car Ratings, which ranks EVs sold in the U.S. based on their green rating using a score from 0-100, considering factors like range, weight, carbon footprint, and battery size. Also, Greenpeace East Asia's Auto Environment Guide 2022 tracks the efforts of the world's ten largest automakers to reduce carbon emissions based on their plans to phase out internal combustion engines, their efforts to reduce carbon emissions throughout their supply chain, and their resource reduction and efficiency measures. Ranking systems such as these, can influence how consumers choose EVs and thus affect the EV market. As a result, automakers are competing to obtain "greener" raw materials which is driving miners to produce greener products.

"Socially responsible" investing has been around for many years, but has recently gained significant popularity. There is a growing trend of individuals with an increasing interest in ethical investing, including retirement funds. It has been estimated that ethical investing may represent about 25% of all managed funds and is an important economic driver. For instance, ethical investing has led to difficulties for some coal mining companies in raising capital. Some investment funds have indicated that they will not invest in carbon-emitting companies in the future.

The credentials of a mining company in good environmental stewardship assist in the social licence to operate within the community. One of the negative impacts of mining on local communities is air pollution and microclimate change. Mine electrification could mitigate this negative impact and may allow providing clean green energy to local communities and nearby townships at a lower cost than existing fossil fuel generators. In addition, through electrification, mining companies can show their commitment to sustainability, potentially attracting environmentally and socially responsible investors and customers.

As a result of the above, major mining companies have made aggressive commitments to reduce their Scope 1 and 2 carbon emissions over the next few decades. Figure 4 illustrates some examples of these ambitious commitments, with all companies aiming to achieve net-zero emissions by 2050 or earlier and having interim targets. The general trends for the proposed emission reductions of the companies shown are similar, but Rio Tinto has a more aggressive earlier target, while Anglo-American has an earlier net zero goal of 2040.

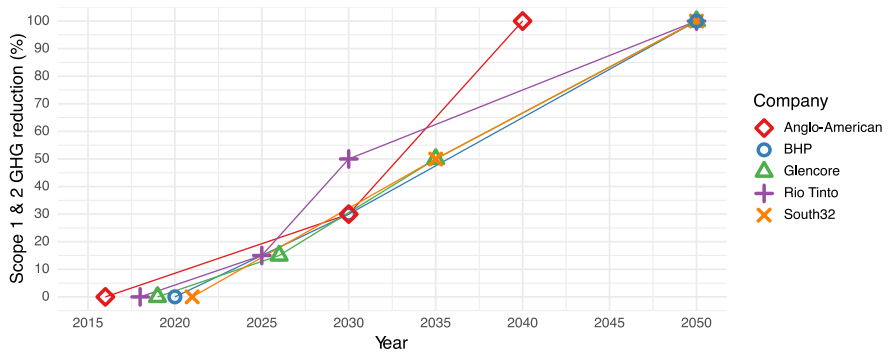

Figure 4: Examples of major mining company commitments to GHG reductions

Subsection 2.2 – Improving Mining Economics

Maximising the financial returns from a mine includes factors such as the cost and time to develop the mine before it starts producing ore and the running costs versus the ore production over the mine's life. In general, transitioning to an electric mining fleet will require higher capital costs. However, it can significantly shorten the development time of the mine (for greenfield sites) and reduce operational costs (for both greenfield and existing sites).

<u>Mine Development</u>

The development time of a mine includes the construction of infrastructure and the installation of necessary equipment and facilities to access the ore body, which can take a number of years. When developing underground mines using conventional diesel vehicles, a significant issue is strict ventilation requirements to remove diesel fumes. This means ventilation shafts must be drilled as the mine tunnels are constructed. Using BEVs (battery-supported electric vehicles) significantly reduces ventilation requirements and hence requires smaller and/or fewer ventilation shafts, which can thus significantly reduce the development time. In some cases, the most significant financial benefits of electrification could be realised during the development phase.

<u>Vehicle Capital Costs and Productivity</u>

Using BEVs instead of conventional diesel vehicles will likely have higher upfront capital costs due to the high cost of batteries and charging infrastructure. In underground mines, haul trucks carry significant

loads of ore (e.g., 50-60 tonnes), usually up steep slopes (typically 10-15% or 4-6 degrees) with hundreds of meters of vertical elevation[1].

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The steep road grades and heavy loads, combined with the need to travel long distances from the loading point to the dumping area, require the vehicle batteries to have both high power capability and high energy capacity. The battery energy requirement could be exacerbated by the limited number of times available to recharge the batteries during a 12-hour shift to meet the productivity targets. Therefore, mining haul trucks are generally considered the most challenging mining vehicle to electrify among the mining fleet. With current lithium-ion batteries, large battery sizes and frequent recharging are required for these heavy duty trucks. Typical options for battery recharging include fast charging, battery swapping, and trolley assist systems. All these charging options require significant upgrades and investment in the electric power system infrastructure to meet the battery energy requirements. In addition, having larger onboard batteries will generally reduce the truck's payload and hence lower its productivity. The charging time also can prolong the time it takes to complete a haul trip, resulting in fewer trips completed during a shift, which further reduces productivity.

One potential solution to improve productivity is to increase the truck's loaded uphill speed, which requires a higher power capability from the electric drivetrain. In addition, careful selection of the battery charging technology can assist in minimising the effect of charging on the vehicle productivity, and in combination with increasing the uphill speed, can even surpass the baseline diesel vehicle productivity. Moreover, future electrified mines may use autonomous haul trucks, and in this situation, a larger fleet of smaller trucks may possibly be the optimal solution.

Mine Operational Costs

Energy, ventilation, and maintenance are the major underground mine operational costs affected by the transition from diesel to BEVs, which is estimated to yield an operational expenditure decrease of 7% to 15%. Energy expenses account for approximately 20% to 33% of total mining operational costs. A fully electrified mine is projected to achieve a 30% to 50% reduction in energy expenses due to the combination of decreased vehicle energy consumption and reduced ventilation costs.

The lower energy costs associated with BEVs can be attributed to their more efficient drivetrain and the ability to use regenerative braking. The increased drivetrain efficiency is associated with the higher efficiency of electric motors compared to diesel engines and the elimination of the transmission. Additionally, regenerative braking is beneficial in mining applications where haul truck operation often includes large vertical elevation changes. BEVs can recover substantial amounts of energy when going downhill, while diesel vehicles convert this energy into waste heat.

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[1] Note that articulated trucks can handle gradients as steep as 12% (1:8 ratio), which is slightly less steep than the common slope of 14.3% (1:7 ratio) for underground vehicles. The slope at each segment is carefully designed taking into account the shape of the deposit, geotechnical factors, and other considerations. Key goals are minimising fuel consumption and reducing safety risks associated with operating on steep slopes.

BEVs have significantly fewer moving parts than diesel vehicles, typically about 25-33% fewer, which should result in reduced maintenance costs. Also, electric trucks provide a greater opportunity for digitalisation, monitoring and measurement, enabling engineers to predict faults in electrical components and improve preventive maintenance to reduce overall maintenance costs further.

Although BEVs likely have lower maintenance costs, they do require regular battery replacement when their storage capacity degrades beyond a certain point. This can represent a significant operational cost. The replacement frequency depends on the amount of energy cycling by the batteries. However, it is possible that these old vehicle batteries can have a "second life" as stationary energy storage units for the mine power system. Any power system based primarily on renewable energy typically requires substantial amounts of energy storage, which can be partially supplied by repurposed vehicle batteries. Furthermore, these batteries could be sold for residential applications in the used battery market.

The diesel particulate matter (DPM) and other emissions produced by diesel engines have been classified as "Group 1 - carcinogenic to humans" by the World Health Organization. The heat, noise, and fumes generated by diesel engines create potentially hazardous conditions in the working environment, especially for underground mining operations. Thus, substantial ventilation energy is consumed in underground mines to maintain the air quality at an acceptable level. This extensive ventilation requires a large network of high-power primary and secondary ventilation fans, which can have total power consumption in the order of megawatts. A significant reduction in the ventilation power requirements is expected by using BEVs. For instance, one case study achieved a 40% reduction in ventilation costs.

An additional benefit of mining vehicle electrification and using renewable energy is the reduced sensitivity of the operational costs to fluctuations in diesel prices and potential cuts in government diesel incentives. Figure 5 depicts the changes in diesel prices in Australia, US and worldwide between 2010 and 2022. The most notable observation is the surge in diesel prices in 2022 by approximately 65% in Australia. A similar trend can also be observed in the United States, where diesel prices rose by 33% in the same year.

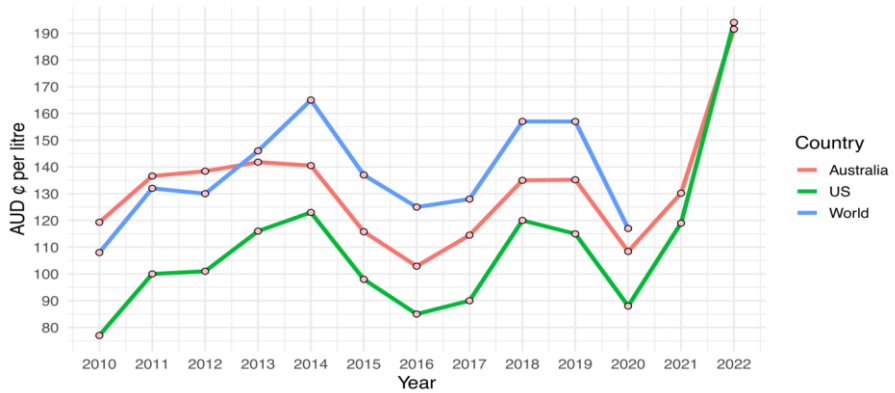


Figure 5: Diesel price variation in Australia, US and worldwide. Note that the worldwide data used here only provides pricing up to 2020.

The electrification of mine vehicles and machinery will likely result in more information being available about the vehicles and how they are used. This provides an opportunity to optimise the use of vehicles to reduce costs. For instance, information about the vehicle battery's state of charge, temperature and state of health will help design more effective charging, discharging and maintenance routines to avoid excessive battery degradation. Additionally, the electrification of mining vehicles provides a pathway toward future automation, offering further opportunities to reduce costs and improve mining safety.

Subsection 2.3 – Mine Worker Health and Safety

Diesel emissions, the exhaust gases of a diesel engine, are a complex mixture of gases and particles released into the air when diesel fuel is burnt. Common components of diesel emissions include carbon monoxide (CO) and dioxide (CO_2), nitrogen oxides (NOx), sulphur oxides (SOx), hydrocarbons (HC), and particulate matter (PM). PMs consist mainly of tiny organic and elemental carbon particles and some sulfuric acid. They can absorb other substances from the environment to become more toxic, and inhaling these can increase the risk of developing long-term health problems, e.g., lung and bladder cancer. It has been estimated that up to 1.2M Australian workers per year are exposed to potentially dangerous levels of diesel particulates. With the electrification of trucks and machinery at a mine site, these underground diesel emissions will be completely eliminated, improving the health and well-being of underground mine workers.

The EURO and US TIER classification system regulates the exhaust emission limits for on- and off-road heavy-duty vehicles. In an Australian mine, most trucks and vehicles that can be driven "on road" must comply with EURO emission limits, while off-road vehicles are regulated by US TIER. These standards, including Euro V (2008), Euro VI (2012), TIER 4i (2011) and TIER 4f (2014), set limits on CO, NOx, HC and PM as a function of engine energy output, with Euro VI and TIER 4f being more stringent. Implementing

these standards has led to improved engines to meet these strict limits. Figure 6 shows the emission limits determined for EURO in a steady test cycle and US TIERs for engine power in the range of 450-560 kW. Note that the amount of ventilation required for an underground mine is determined by the legislation based on the emission specifications of a diesel vehicle.

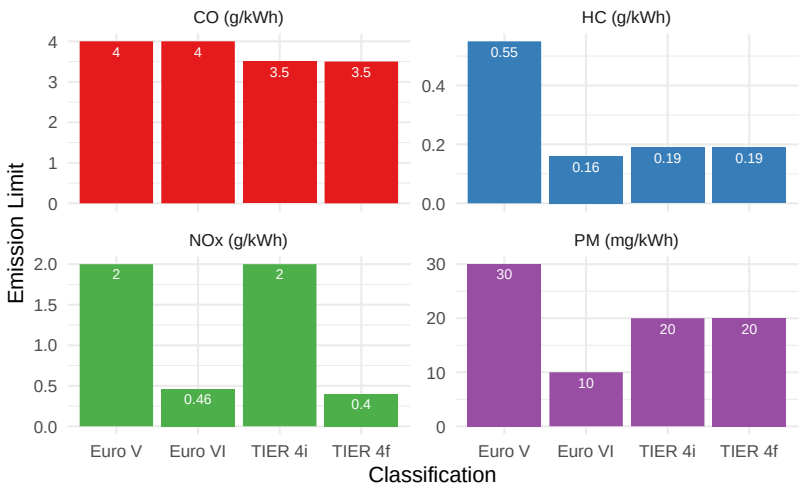


Figure 6: Comparison of EURO and US TIER emission standards for on-road and off-road heavy-duty vehicles.

In addition, compared to diesel vehicles, BEVs generate roughly only one-eighth of the waste heat, which can enhance the feasibility of new projects in deep mines associated with active geothermal conditions by mitigating the heat factor.

Last but not least, BEVs are significantly quieter than diesel vehicles. High levels of acoustic noise can occur with diesel vehicles due to their high-power engines operating in confined spaces. In early BEV trials, mine workers indicated that reduced noise levels were an important benefit of these vehicles.

Section 3 – Three Essential Steps for Effective Underground Mining Electrification

A successful transition to underground mining electrification necessitates a systematic approach that includes a comprehensive cost-benefit analysis while carefully assessing potential risks. For a new underground mine (greenfield project), the design for electrification should begin at the inception of the project. This allows for consideration of the placement of charging infrastructure, selection of the electric fleet, and design of the mine specifically for the electric fleet, taking into account its unique characteristics and differences. On the other hand, an operational mine (brownfield project) faces the challenge of retrofitting. Strategizing how to gradually shift toward electrification requires blending old approaches with new ones. Existing infrastructure, operational practices and schedules, and machinery might need modification or replacement. Moreover, maintaining production while ensuring safe and efficient electrification demands careful attention. We propose three essential steps that cover the electrification needs for both brownfield and greenfield mines from a high-level perspective: data collection, optimal system design, and workforce development. These steps aim to ensure a technically and financially viable transition to electric fleet operation. Data collection lays the groundwork for designing an optimal system, which subsequently informs the necessary workforce training to operate and maintain the newly implemented electric fleet. These are explained in detail in the following subsections.

Subsection 3.1 – Data Collection

The first step towards transitioning to an electric fleet involves gathering data on mobile and fixed machinery, as well as mining operations, to determine the electrification status quo of the mine for both existing and new mine sites. To quantify the benefits of transitioning, the electrified mine can be benchmarked against the baseline mine equipped with a diesel fleet. A substantial history of operational data is available for an existing site for calculating benchmarks for comparison. However, some data may not have been measured or recorded, leading to gaps in data collection. These gaps must be filled with assumptions, potentially compromising an accurate understanding of the factors needed for optimal design. Therefore, a deliberate and detailed plan is essential to measure and collect the appropriate data for a sufficient duration and sampling frequency before implementing electrification.

Economic factors such as capital, operational, and maintenance costs contribute to a realistic cost analysis for the mine. These economic factors include the related costs of diesel vehicles, ventilation systems and electricity generation plant. Moreover, other technical and financial parameters necessitate knowledge or predictions of future mine operations and expansion, such as potential changes in haul profiles in the future.

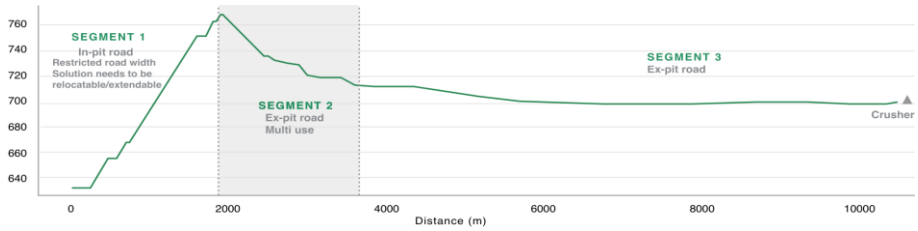
Based on our experience from the MOVE project, we categorise the required data into five groups for an existing underground mine:

1. Mining fleet: Haul trucks play a key role in mobile fleet data collection, as previously mentioned, they pose significant challenges when converting to BEVs. Nevertheless, the entire vehicle fleet

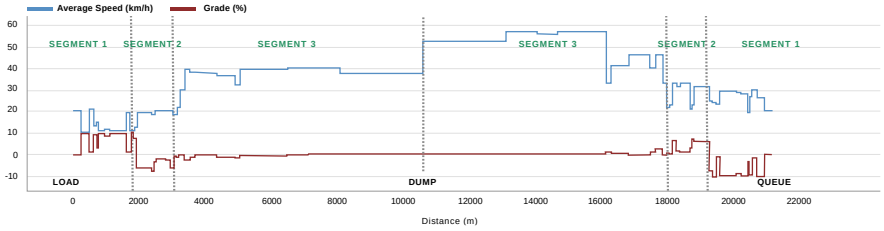
must be considered during data collection. This category includes information and specifications on the current mining vehicle fleet, usage patterns, maintenance schedules, fuel consumption data, and unexpected downtime rates.

2. Haul profile: Assessing the energy requirements for the operation of BEV haul trucks relies on material movement data from production to the dump point, which defines how much energy and power is required for the electric fleet. This category includes material movement rates, haul profile specifications, maximum vehicle speed in different segments, information on the potential implementation of battery charging technologies and statistical data on the history of the material movement for different haul profiles. Figure 7 shows sample data on haul profile specifications obtained from the Charge-On Innovation Challenge. The engine load refers to the power of the truck delivered by the engine.
3. Ventilation system: Primary and secondary ventilation fans are significant consumers of electricity. Figure 8 illustrates a simplified ventilation system for an underground mine, indicating the inlet and outlet air pathways relative to the mine structure. The primary ventilation is responsible for circulating air through the main tunnels of the underground mine, while secondary ventilation ensures the necessary airflow within side tunnels. So, a comprehensive understanding of the ventilation configuration and its energy consumption is essential for determining the extent to which ventilation can be reduced due to vehicle electrification. Other important factors in this category encompass anticipated future expansion, specifications of the primary and secondary fans, as well as ventilation management guidelines.
4. Electricity infrastructure: An electric mine with a sophisticated charging infrastructure for the smooth operation of an electric fleet requires a thorough understanding of the infrastructure and the reinforcement of the existing underground electric power system network in terms of capacity, reliability, and safety. This category includes a high-level single-line diagram for the power system, generation sources, power system arrangement (owned or contracted), current or proposed power purchase agreements and electricity demand time series data at the individual feeder level.
5. Power supply: Besides load management, comprehensive information on the power supply is required to understand available sources and evaluate potential future power sources. This category can be divided into several groups as follows:
    a. Regional electrical grid connection: When a connection to a regional power grid is available, we need to know its ratings, the tariff structure, the reliability and power quality metrics, and opportunities to reinforce the connection.
    b. Fossil-fuel electricity generation: Specifications of existing fossil-fuel powered generators, annual load profile, operating cost, and emissions data.
    c. Renewable energy: System specifications, resource assessment, historical generation profile, and environmental considerations.
    d. Energy storage: System specifications, historical operating profile, opportunities to add additional energy storage systems, a list of feasible energy storage resources in the

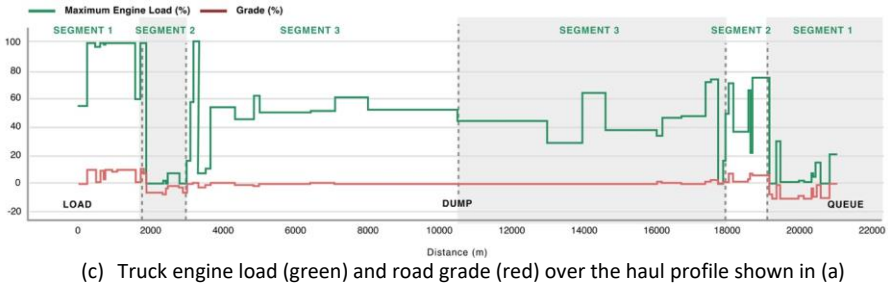
locality (e.g., small pump hydro, air storage, etc.), environmental conditions (to suit operational limitations of certain storage technologies), CAPEX and OPEX, outages, degradation (if relevant) of the existing storage units.



(a) A typical open-pit haul truck profile – obtained from the Charge-On Innovation Challenge



(b) Truck average speed (blue) and road grade (red) over the haul profile shown in (a)



(c) Truck engine load (green) and road grade (red) over the haul profile shown in (a)

Figure 7: Sample data for mine electrification application

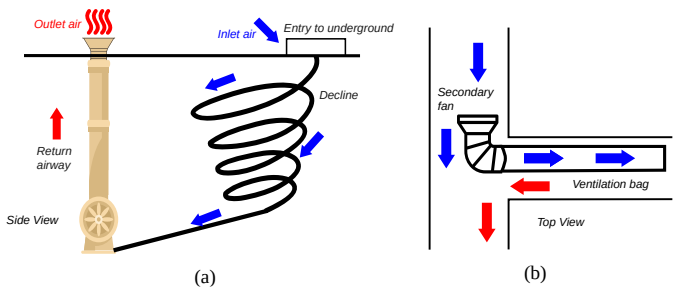


Figure 8: Sample illustration of the ventilation system with primary (a) and secondary (b) ventilation in an underground mine

Subsection 3.2 – Optimal System Design and Operation

The system design of a mobile electric fleet is a multifaceted and multi-disciplinary problem encompassing the complexities of electric fleet scheduling for smooth and efficient operation and load and supply management for the electricity infrastructure. Electrification will require a significant upfront investment, but should reduce emissions and operating costs and improve safety. In the following, we highlight three important aspects of the systemic design and operation of the electrified mine site.

Rethinking mine design and operation: It is necessary to re-evaluate conventional mine designs that have been used for diesel fleets for decades, and adapt operations for BEVs. As mentioned earlier, this may allow significant reductions in the time spent on developing the mine. Another area is determining the optimal tunnel slope for BEVs, which involves balancing the cost and time required for construction against the vehicle drive power required for propelling the fully-loaded truck up the slope. Electrification can also be regarded as a stepping stone to underground vehicle digitalisation and autonomous operation. BEVs can provide more detailed real-time monitoring of vehicle performance, such as truck speed, battery state of charge, and temperatures, thanks to a large variety of onboard sensors. For autonomous trucks in future mines, with the elimination of labour costs and constraints, the optimal haul truck size may likely be different. This could, for instance, involve using a larger number of smaller vehicles to reduce tunnel dimensions, and hence saving infrastructure expenses and shortening development time.

Whole-of-system design and control: The electric power system works in a profoundly interconnected fashion. For instance, ventilation is a major energy consumer in underground mines, and the amount of ventilation needed to comply with legislation depends on the locations and operation of underground vehicles. These vehicles have onboard batteries which need to be charged regularly and form another

significant load on the power system. When considering renewable energy sources, fluctuations exist in both PV and wind power generation. Designing each subsystem of the energy infrastructure independently requires storage of large amounts of energy to compensate for short, medium, and long-term variability in their generation. This oversizing issue and power supply instability can alternatively be addressed by actively controlling the BEVs' battery charging and, in some cases, perhaps discharging these batteries to supply energy back to the power grid (vehicle-to-grid, V2G). Hence, adopting a whole-of-system design is essential to achieve optimal cost reduction, reliability, and seamless operation of the entire mine. So, the whole-of-system design can identify interdependencies, synergies, and potential bottlenecks between the energy supply (e.g., renewables, energy storage, or backup generators) and the onboard battery and charging infrastructure and enable better decision-making with a more effective system design.

Flexibility for future mine expansion: Whether it is an existing or new mine, an optimal system design must include provisions for future mine expansion plans. Although it is challenging to predict the mine's evolution over its lifespan, adopting a modular and smart design could help reduce the future costs. This is particularly significant in electrified mines because battery charging requires significant infrastructure. Regardless of the battery charging technology used (e.g., fast charger, battery swapping, trolley-assist charging, wireless charging, or a combination of these), the locations of these charging points can be an important cost factor during the mine's evolution. In addition, adopting a modular design for the charging infrastructure can offer a flexible, reusable, and scalable solution that can be adapted to the mine's changing requirements over time. Once a mine becomes operational, ventilation requirements can change as the mine expands both vertically and horizontally. This expansion can create areas with low airflow. To address this, additional ventilation infrastructure, such as auxiliary fans, new ventilation drifts, or even shaft deepening, might be necessary. This could require a re-routing of the mine's ventilation electricity network and an extension of the electrical system infrastructure. Therefore, these factors must be considered flexibly in the planning of the mine's electrical system.

Subsection 3.3 – Future Workforce Training

The mining industry is undergoing a significant transformation fuelled by electrification, automation and digitalisation. These technologies are expected to reshape the industry and create new opportunities for the workforce while raising challenges in adjusting to new work practices. Opportunities include increased safety, improved efficiency, and new roles in technology. However, the challenges include learning to operate and maintain BEVs and the ability to work effectively with increasing amounts of data. The future workforce will need to operate in a highly integrated human-machine setting, where machines often support decision-making. There is a growing reliance on remote operations and digital presence.

A recent EY report predicts an increase in employment in the mining sector over the next five years and that almost 75% of the occupations in the sector will be enhanced or redesigned due to new technologies. Among the skills in demand are system evaluation and analysis, mathematics, data

analysis, data and digital literacy, judgment, and decision-making. Conversely, the skills with decreasing demand include vehicle operation, materials extraction, operations and control, equipment maintenance and blast-hole drilling.

In the following, we highlight four categories of emerging skills that require re/upskilling for the future workforce in preparation for the upcoming changes.

Data and digital literacy: Mine electrification enables the real-time collection, analysis, and storage of vast amounts of operational data ranging from ore grades to equipment performance. This data can help mining companies identify areas for improvement and optimise their operations accordingly. By doing so, they can increase efficiency, lower costs, and enhance worker safety. Advanced data analytics empower miners to make well-informed real-time decisions to improve overall mining operations. To achieve this, mine designers and operators must collect the appropriate data and leverage digital literacy to use the data. In addition, smart and digital mining entails the use of an interconnected network of devices, machines and sensors to collect and exchange data based on Internet of Things (IoT) architectures, which can be used to monitor equipment and gather real-time data on operations. OEM engineers developing innovative mining equipment and processes must possess data and digital literacy so they can utilise digital technologies such as real-time simulation and artificial intelligence alongside extensive field data to test and refine new designs before production. This approach helps reduce the time and cost of introducing new equipment to the market while also improving the reliability and performance of the equipment.

Equipment maintenance and operation: Electrification requires reskilling the workforce engaged with diesel vehicles to ensure their adaptability and competence in handling BEVs. The shift to electrification presents an opportunity to close the skill gap by providing targeted training programs to empower workers with the necessary new skills to maintain, repair and operate BEVs. These programs should cover a deep understanding of the new equipment and its components, including battery, thermal management systems, battery management system, power electronic converters, electric motors, high-voltage DC circuits, etc. These emerging skills are notably distinct from the existing skill set required for diesel vehicle maintenance.

Different approach to training the future workforce: The accessibility to digital twins of mine operations will grow as mining electrification advances. This provides an opportunity to employ virtual reality (VR) for training the workforce on mining equipment and operations. This revolutionary shift calls for developing a new generation of training materials and software packages. The adoption of VR requires the industry to embrace a new mindset and cultivate a workforce comfortable with learning in virtual environments and engaging in human-machine interactions.

New safety training and regulations: The emergence of new hazards associated with EVs in mining operations underscores the importance of reskilling and upskilling the workforce to adapt to these challenges. The Australasian Fire and Emergency Service Authorities Council (AFAC) released a report in

2023 regarding incidents involving EVs which outlines a variety of hazards related to battery operation, such as thermal runaway, the danger of high voltages, and the fire intensity at higher battery state of charge levels. As a result, the New South Wales Resources Regulator has outlined a list of safety recommendations that mine operators should implement in the case of a lithium-ion battery fire, including emergency management plans, first responders trained in the use of personal protective equipment (PPE) and understanding of the risks associated with the use of lithium-based batteries in mines, new ventilation requirements for battery electric equipment operation, and ventilation for underground charging stations. This calls for a range of new skills and training for the first responders, machine operators, and other workers near BEVs, and also that the mine designer needs to carefully consider these requirements in designing and/or expanding a mine site.

From a global perspective, the landscape of EV regulations and standards remains varied. Presently, no single global EV standard exists, with major EV production hubs such as Japan, Europe, North America, and China each advocating for distinct regulations in multiple domains.

The United Nations Economic Commission for Europe's Global Technical Regulation No. 20 aims to harmonize electric vehicle regulations worldwide. This regulation addresses the potential safety risks of EVs during operation and post-crash scenarios, focusing on electrical shocks from high-voltage circuits and the inherent hazards of lithium-ion batteries or other rechargeable electric energy storage systems. In parallel, CharIN (Charging Interface Initiative e.V.) has recently launched a Mining Taskforce in partnership with the International Council on Mining and Metals (ICMM). The goal of this taskforce is to facilitate the electrification of the mining industry and support the transition to electric mine haul trucks while enabling increased operational efficiency, reliability, and safety.

Conclusions

This article highlights the importance of mine electrification and the merits of using heavy electric fleets for material transportation in underground mines. We discuss the challenges associated with the transition towards electrification and summarise the essential steps for practical underground mining electrification.

The ongoing transformation towards electrifying haul trucks and mining equipment is driven by economic, environmental, social, and governance aspects, which have been discussed in the article. The expected outcomes of this transformation include higher operational efficiency, a boost in sustainability, and increased safety in both underground and open pit mines. Moreover, implementing emerging technologies such as autonomous vehicles, digitalisation, IoT-based communication and big data analytics can improve productivity and decrease costs throughout the supply chain, from material extraction to transportation.

While mine electrification holds many benefits, the transition to the electric mining process is complex and multifaceted. It entails developing new technologies or repurposing existing ones, investing

significantly in infrastructure and hardware, modifying processes, and training the workforce with new skills. If done improperly, this could lead to higher operating costs and reduced productivity. This article outlines three necessary steps based on our first-hand experience with a large-scale mining electrification project initiated in 2021. Developing site-specific practical solutions, including modelling and assessing real-world case studies, has given us invaluable insights into providing a tailored guide to fit-for-purpose electrification solutions ranging from battery sizing and selection to charging infrastructure design to assist battery equipment, technology, and service providers to service mining companies during the transition.

For Further Reading

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